Tissue Repair: Wound Healing

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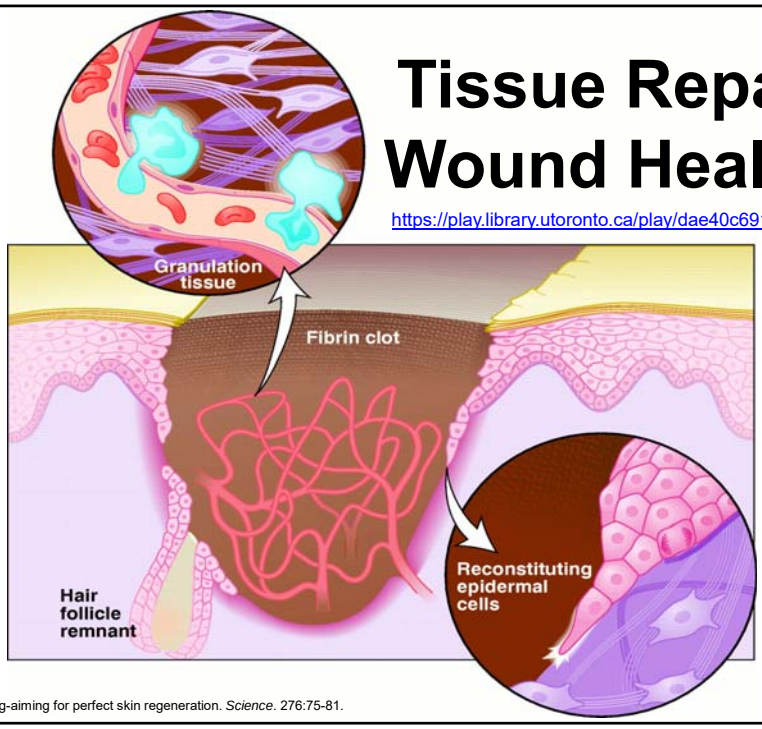
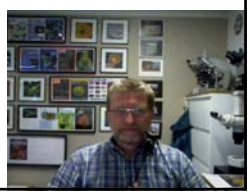


Please use the above link to watch the lecture.
There is an odd pause between 26:50 – 29:41.
Probably use it to reflect on the meaning of the universe and all.

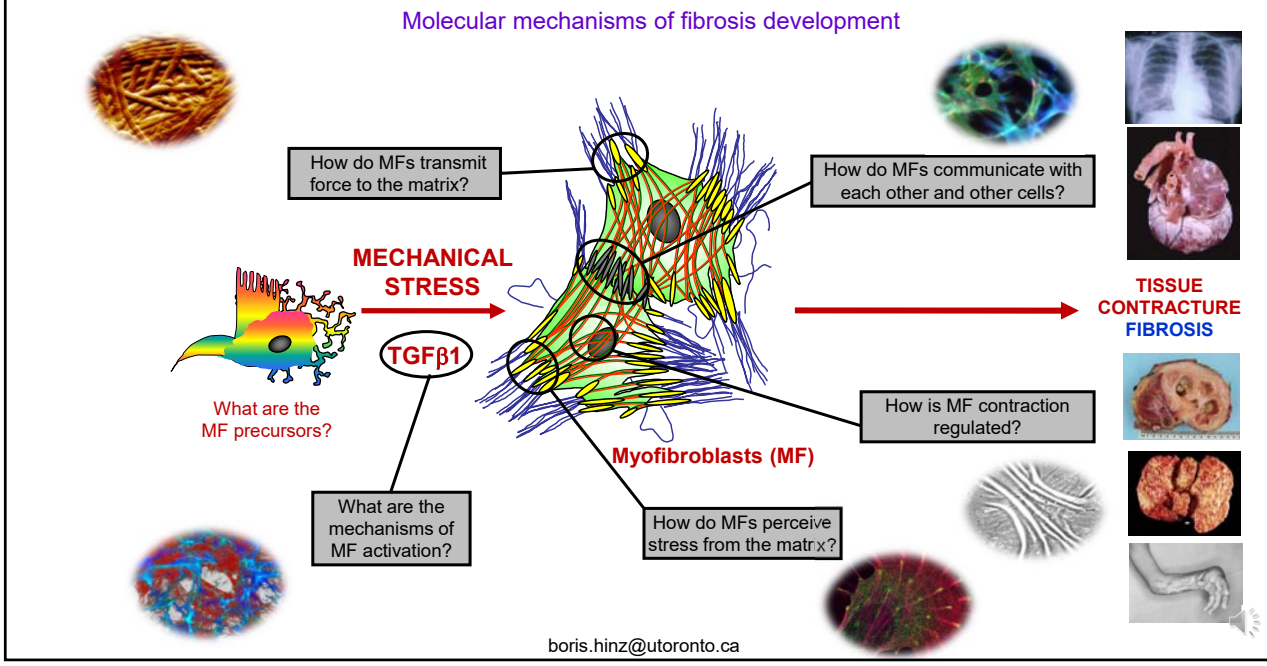


Martin, P. 1997. Wound healing-aiming for perfect skin regeneration. *Science*. 276:75-81.

1

Laboratory of Tissue Repair and Regeneration

Molecular mechanisms of fibrosis development



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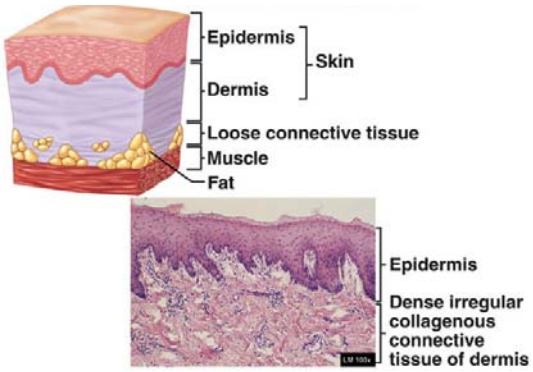
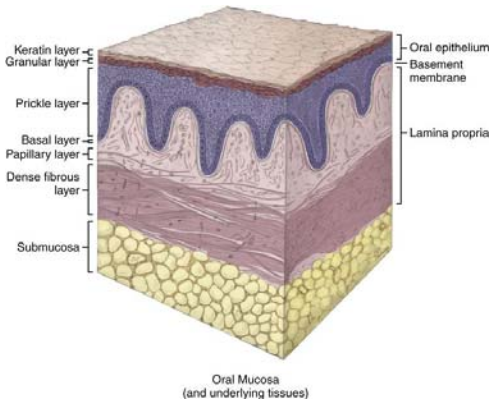
2

The Players in Skin Wound Healing

Adult skin consists of two major layers:

- A stratified epithelium, the epidermis, supported by an underlying
- connective tissue, the dermis.
- Both tissues are separated by the basement membrane.
- Wounding disturbs this balanced arrangement.





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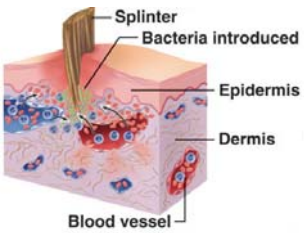
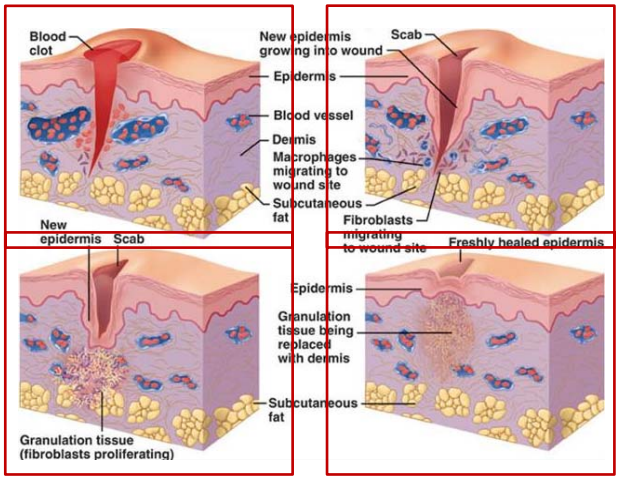
3

Wound Healing is a Multi-Step Process





Wound healing consists of three overlapping phases:

- Inflammatory Phase (Reaction)
  - Hemostasis
  - Phagocytosis
- Proliferative Phase (Regeneration)
- Maturation Phase (Remodeling)

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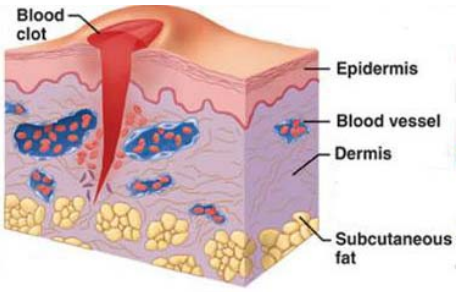
Inflammatory Phase - Hemostasis

- Hemostasis aims at the cessation of bleeding.
- It results from vasoconstriction of the larger blood vessels in the affected area, retraction of injured blood vessels, deposition of the extracellular matrix protein fibrin and formation of a blood clot.

- The blood clot includes blood platelets surrounded by a provisional matrix of fibrin fibers and ECM proteins vitronectin, plasma fibronectin and thrombospondin, providing the framework for repair.
- The clot also serves as a reservoir for growth factors and cytokines, released from degranulating platelets.
- A scab forms at the surface to inhibit contamination by microorganisms.
- The blood supply increases causing redness and edema.



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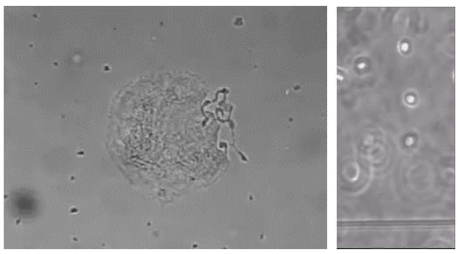
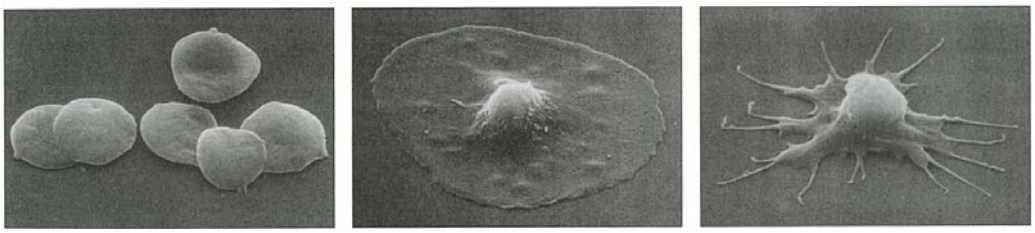
Hemostasis – Platelet Production

- Megakaryocytes are the precursors of platelets.
- Platelets are formed by pinching off the giant megakaryocyte and are released into the blood stream.
- Physical contact with a damaged vessel or chemical signaling through thrombin triggers a signaling cascade in the activated platelet that reacts by spreading.



Platelet activation *in vitro*



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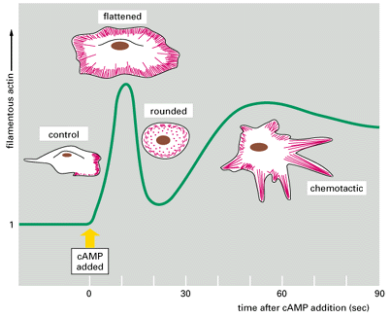
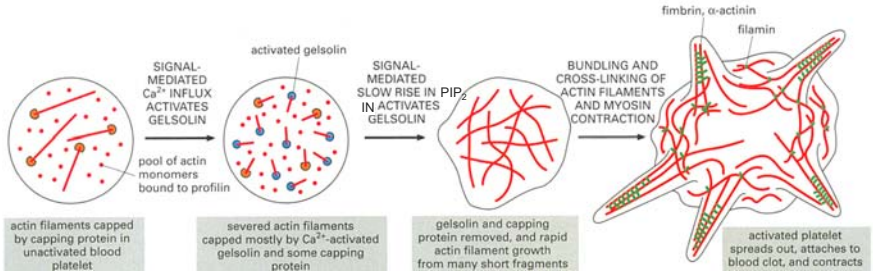
Hemostasis – Platelet Activation

- Spreading is triggered by a rapid Ca^{2+} influx that activates severing actin binding proteins.
- The slower generation of PIP_2 inactivates severing actin-binding proteins, leading to actin polymerization.





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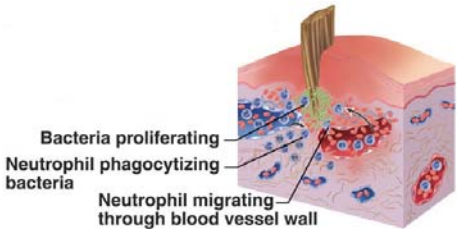
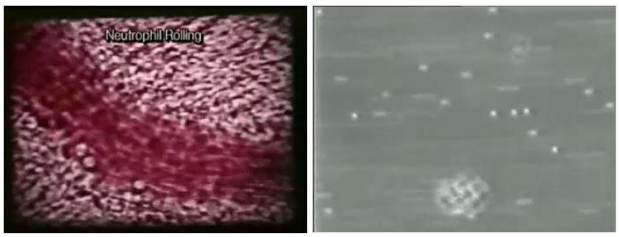
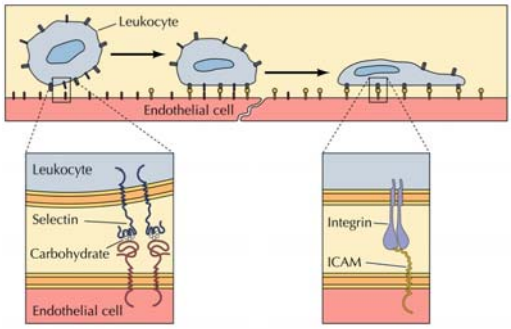
Inflammatory Phase - Phagocytosis







- Vascular and cellular responses occur to remove foreign substances and dead cells.
- Vessel endothelial cells at the infection site express specific surface receptors.
- Leukocytes (neutrophils) adhere to these surface proteins and slow down by 'rolling' on the endothelium

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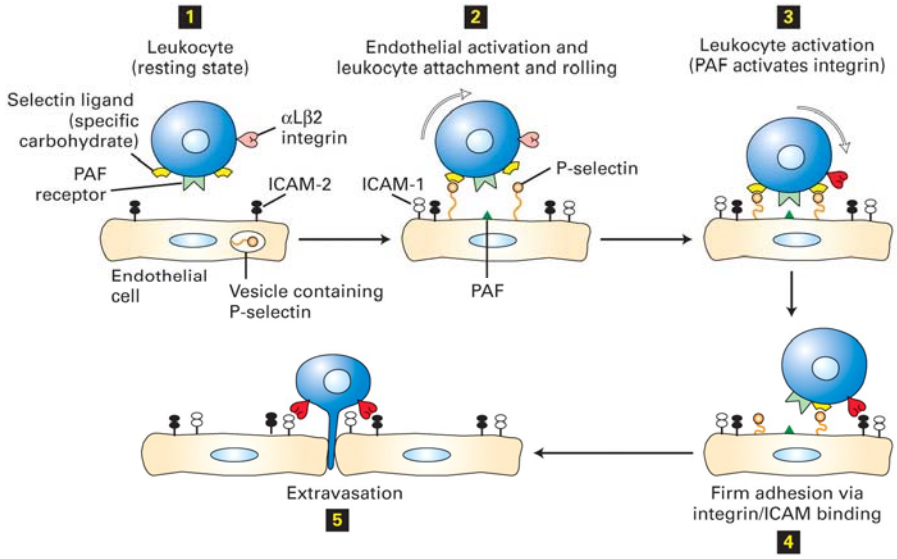
Inflammatory Phase - Phagocytosis



PAF - platelet activating factor

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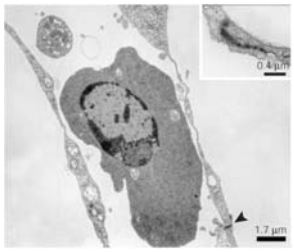
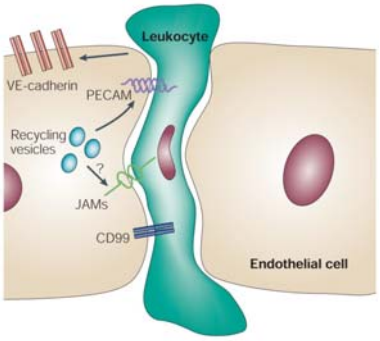
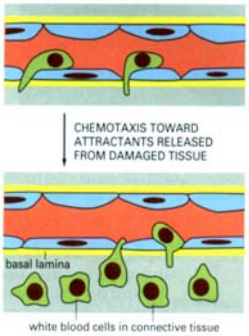
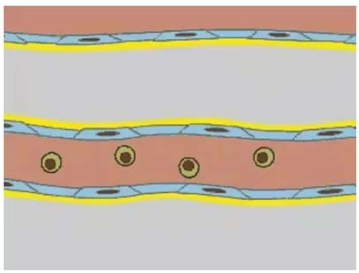
Inflammatory Phase - Phagocytosis

- Once slowed down, neutrophils engage integrins that increase cell adhesion strength.
- Firmly attaching cells leave the blood stream and invade the interstitial space by 'squeezing' between endothelial cells.
- This process requires disassembly and re-assembly of cell-cell adherens junction.









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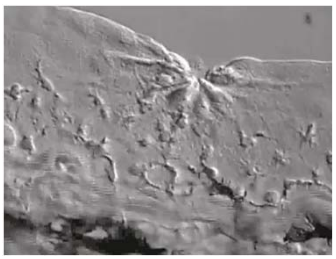
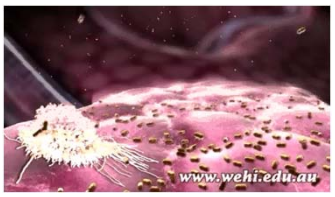
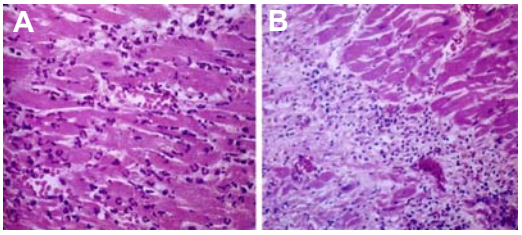
Inflammatory Phase - Phagocytosis

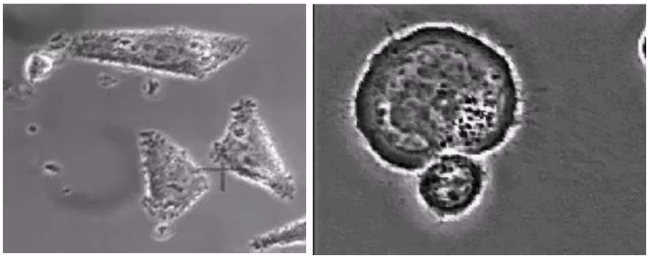
- After ~24 hours, neutrophils are replaced by macrophages.
- Macrophages engulf microorganisms and debris by phagocytosis.
- Macrophages secrete factors, which stimulate formation of endothelial buds at the termini of injured blood vessels.







Immune cell infiltration of infarcted myocardium:
A: early response by neutrophils
B: later response by monocytes and macrophages



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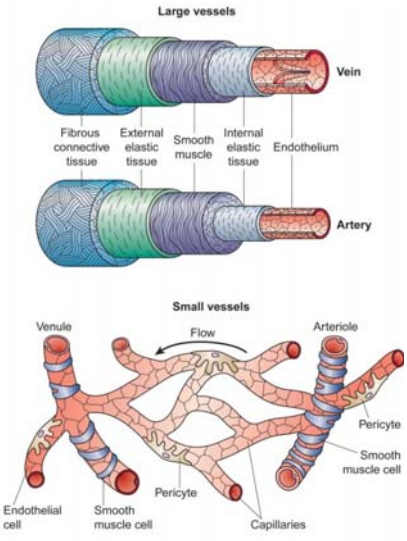
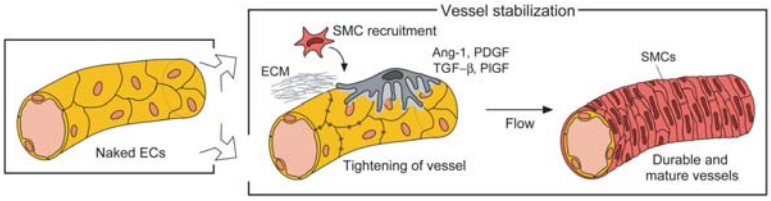
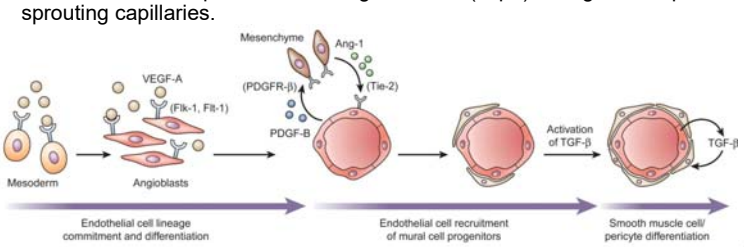
Proliferative Phase – Re-Vascularization





- The wound is invaded by numerous capillaries, resulting in a granular tissue appearance = granulation tissue.
- FGF2 and VEGF, released by damaged endothelial cells, macrophages and keratinocytes (VEGF) promote angiogenesis.
- Endothelial cells express a new integrin subset (avβ3) to migrate at tips of sprouting capillaries.



Cleaver, O. 2003. *Nat Med*. 9:661. Carmeliet, P. 2003. *Nat Med*. 9:653.

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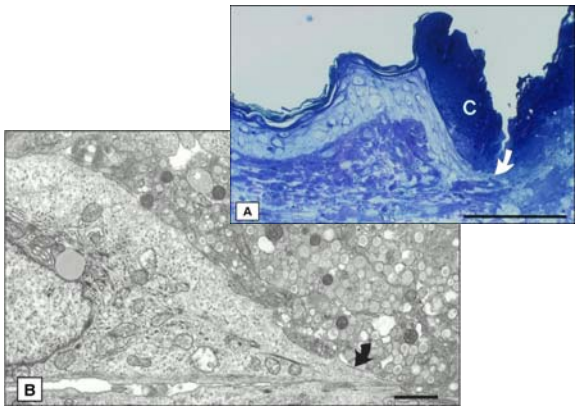
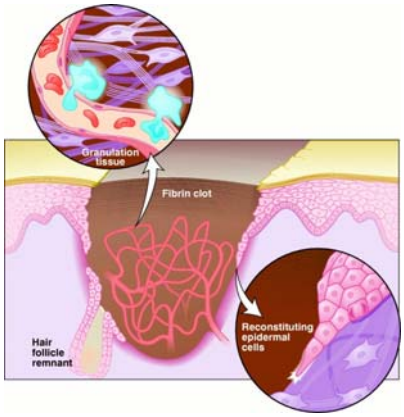
Proliferative Phase – Re-Epithelialization

- Under the scab, epithelial cells migrate into the wound from the edges.
- If wound closure does not occur this way, the denuded area is covered with dried plasma protein and dead cells: eschar.
- If the wound is not re-epithelialized, it becomes covered with thick, gray, fibrous tissue that is eventually converted into scar tissue.



Keratinocyte migration in wounded cornea



Martin, P. 1997. *Science*. 276:75-81.

Zhao, M. et al. 2003. *Faseb J*. 17:397-406

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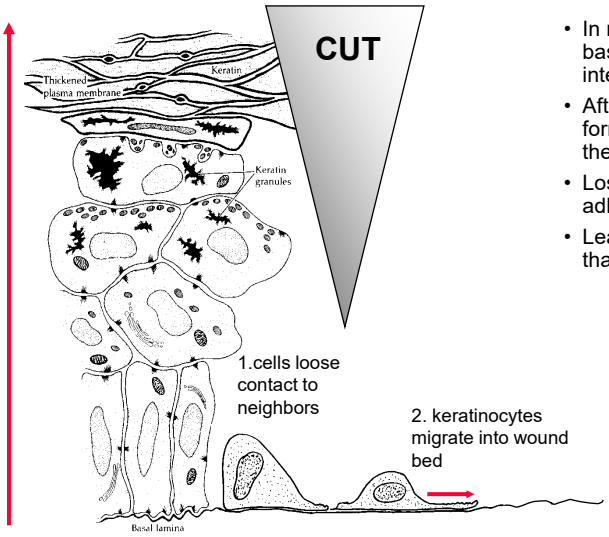
Proliferative Phase – Re-Epithelialization



- In normal epidermis, keratinocytes are anchored in the basement membrane at hemidesmosomes, linked to intermediate filaments.
- After wounding, cells acquire a migratory phenotype and form actin-associated adhesion structures; they change their integrin expression pattern.
- Loss of cell-cell adhesion and change in cell-matrix adhesions correlates with changes in IF organization.
- Leading keratinocytes express proteases (e.g. plasmin) that facilitate clot invasion.

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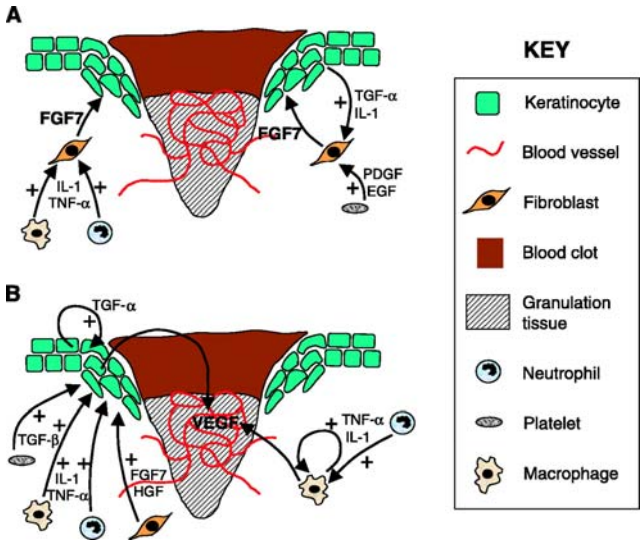
Growth Factors Regulate Regeneration



Important growth factors (GFs) in wound healing:

- VEGF – vascular endothelial GF
- EGF – epidermal GF
- FGFs – fibroblast GF family
- NGF – nerve GF
- KGF - (FGF7) – keratinocyte GF
- TGFα – transforming GF a
- TGFβ – transforming GF β
- Activins (TGFβ family)
- HGF – hepatocyte GF / SF
- PDGF – platelet-derived GF
- TNF – tumor necrosis factor
- IGF – insulin-like GF
- CTGF – connective tissue GF
- ILs – interleukin family
- GM-CSF – granulocyte-macrophage colony stimulating factor

Werner, S. 2003. *Physiol Rev*. 83:835-70.

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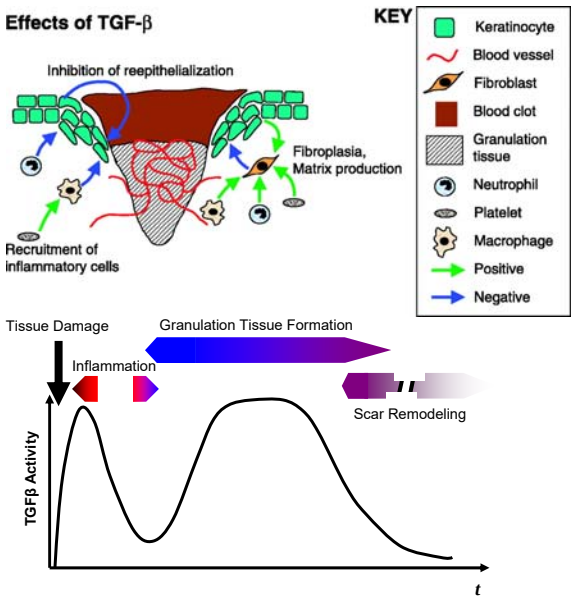
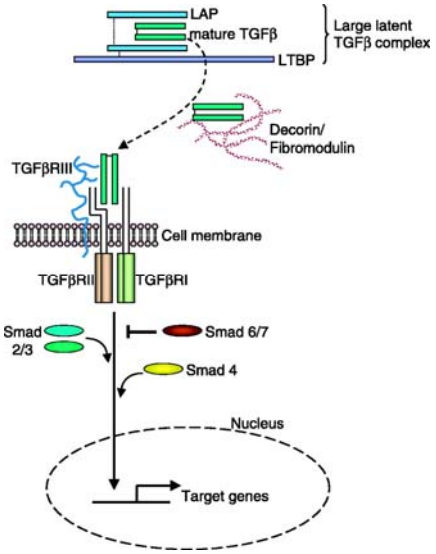
TGF-β – a Multipotent Growth Factor





Werner, S. 2003. *Physiol Rev*. 83:835-70.

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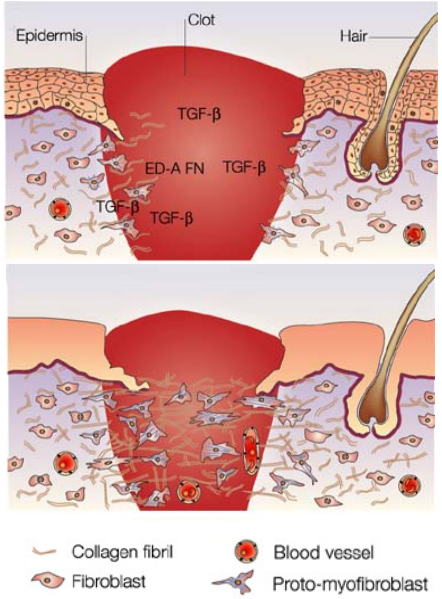
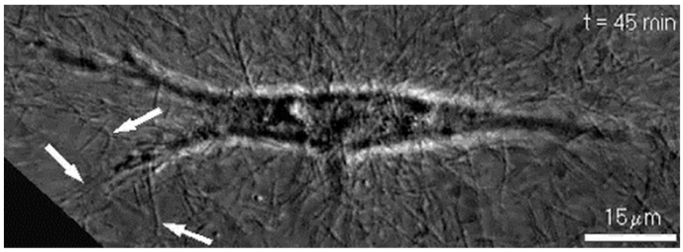
Proliferative Phase – Fibroblast Migration



- Regeneration occurs from day 3-2 to day 21 post-injury.
- Fibroblasts migrate to the wound within 24h and begin to synthesize collagen.
- Collagen adds tensile strength to the wound.
- Capillaries grow across the wound, increasing blood supply.
- Tissue becomes translucently red; at this stage it is fragile and bleeds easily.

Migrating fibroblast in collagen gel



Tomasek et al., Nat Rev Mol Cell Biol 3, 2002, 349-363

Petroll, W.M. et al. 2004.. *Scanning*. 26:1-10.

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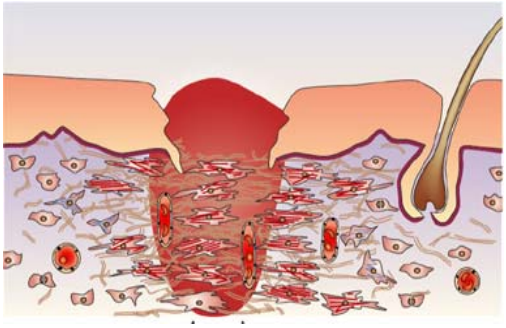
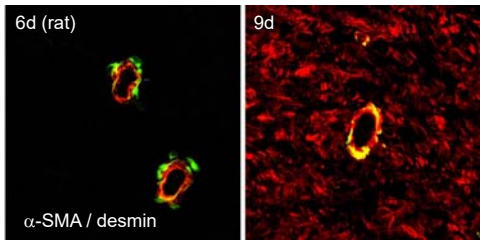
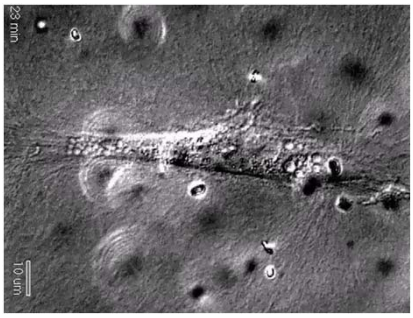
Maturation – Granulation Tissue Contraction

- Begins about day 21 and may last 1-2 years after injury.
- Fibroblasts continue to synthesize collagen, which gets organized into a more orderly structure.
- Increasing tension in conjunction with TGFβ and specific matrix proteins (ED-A FN) promotes myofibroblast differentiation.
- Wound is remodeled and contracted.





Contracting fibroblast in collagen gel



Hinz, B. et al. 2001b. *Am J Pathol.*

Petroll, W.M. et al. 2004. *Scanning*. 26:1-10.

Tomasek et al., Nat Rev Mol Cell Biol 3, 2002, 349-363

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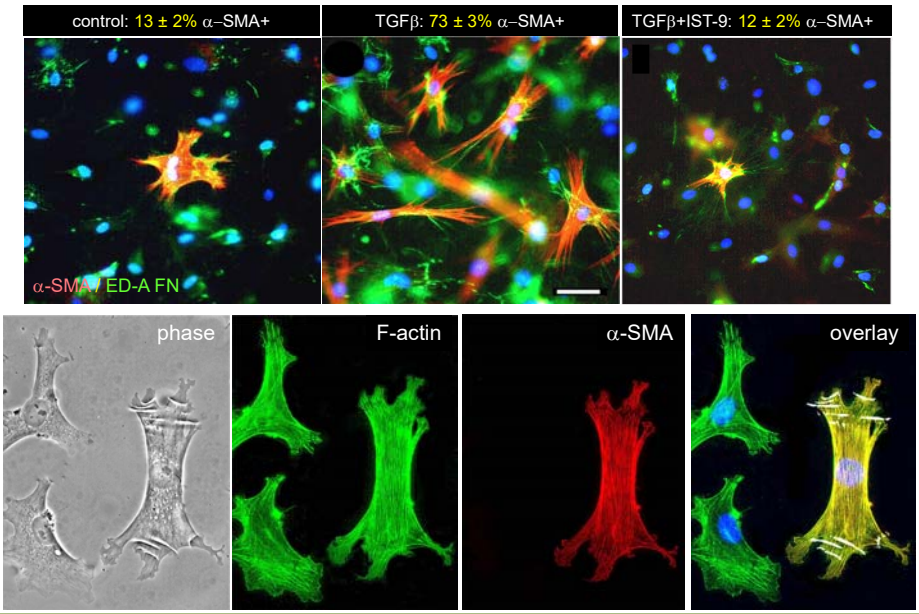
Wound Contraction - Myofibroblasts



Hinz, B. et al. 2001. *Mol Biol Cell*. 12:2730-41.

Slide 19

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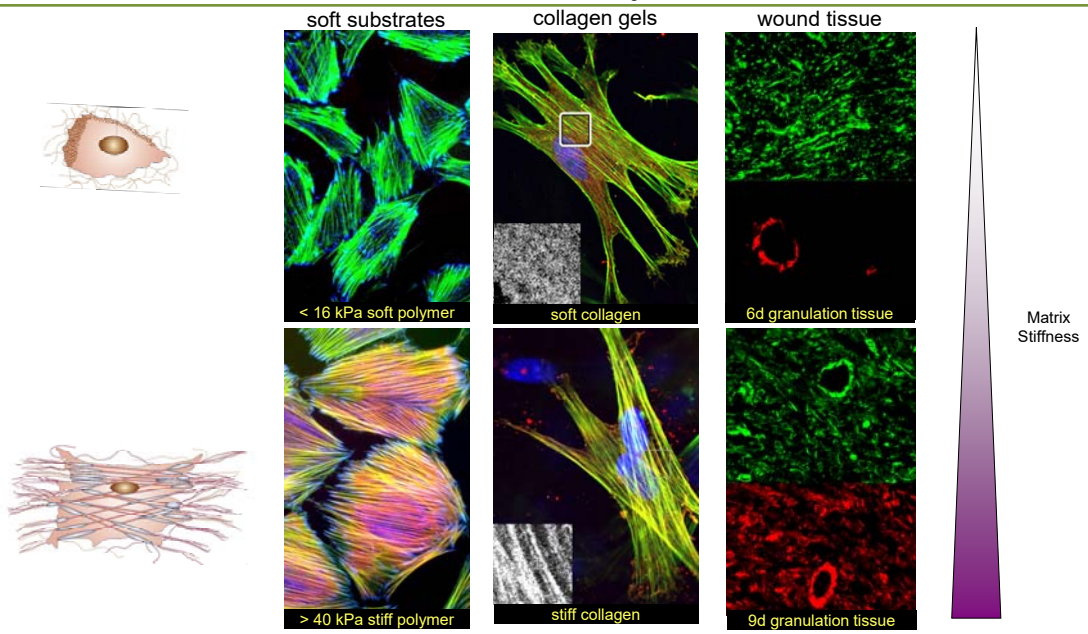
Matrix Stiffness Induces Myofibroblast Activation



α-SMA F-actin vinculin

Hinz, B. (2006). Eur J Cell Biol *85:*

Hinz et al. (2001), Am J Pathol 159

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Tensional Homeostasis (3D collagen)

Tensional Homeostasis in Dermal Fibroblasts: Mechanical Responses to Mechanical Loading in Three-Dimensional Substrates

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[2]Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts
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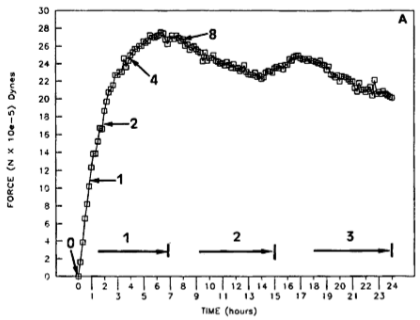
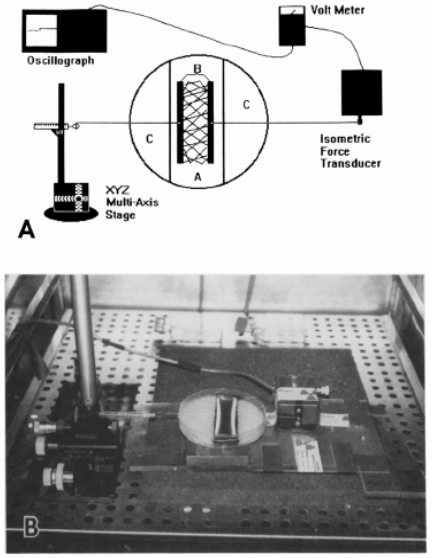
COLLAGEN GEL CONTRACTILE FORCES





Brown et al., J Cell Physiol 175: 323-332

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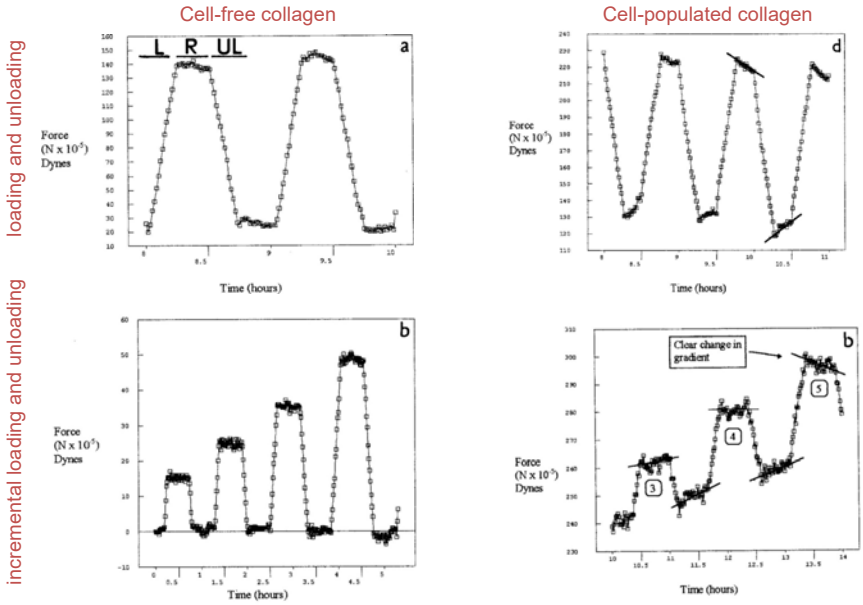
Tensional Homeostasis (3D collagen)



Brown et al., J Cell Physiol 175: 323-332

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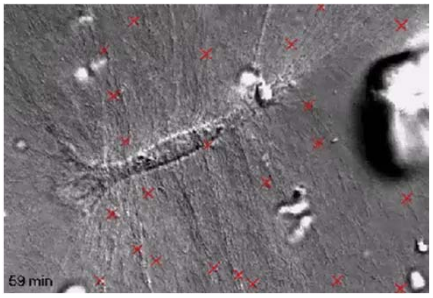
Local Tensional Homeostasis (3D collagen)

Collagen gel compression:

Collagen gel compression + cell relaxation:

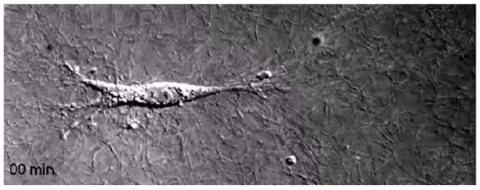


Compression:
- Initial response: Contraction.
- Secondary response: Spreading.

Compression and Cell relaxation:
- Inhibition of cell contraction and depolymerization of F-actin abolish both, the primary and secondary response to ECM compression.

Stretching:
- Cell retract and then re-spread
- During re-spreading no traction occurs

Collagen gel stretching:



Petroll et al., Invest Ophthalmol Vis Sci. 2004 Oct;45(10):3466-74

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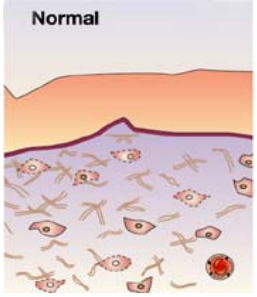
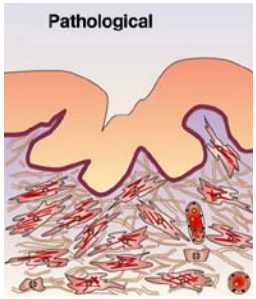
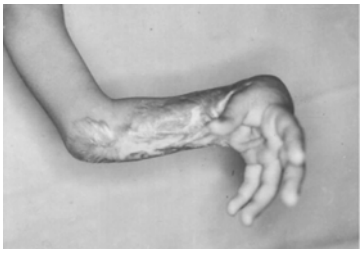
Fibrosis - Connective Tissue Contracture





- After normal wound healing, myofibroblasts disappear by apoptosis.
- Continuous collagen secretion, remodeling and contraction leads to connective tissue contracture.
- Examples: Hypertrophic scars, keloids, fibrotic diseases.



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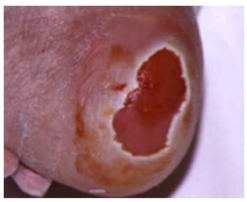
The Healing Problem: Too Little or Too Much?

Chronic Healing	‘Scarless’ Healing	Hypertrophic Scarring
		
Ischemia, Hypoxia Neuropathy Infection - Biofilm Chronic Inflammation Growth Factor Imbalance Lack of Extracellular Matrix Lack of Mechanical Tissue Stress	Everything just right	Hypoxia/Hyperoxia? Chronic Inflammation Growth Factor Imbalance Excessive Extracellular Matrix Contracted and Stiff Scar Tissue

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